

II. PILLAR I — THE ENVIRONMENTAL INPUT

a. Introduction to Pillar I

This pillar examines the first stage of the compound pattern: the environmental conditions and exposures the Government of Alberta either permits, has not regulated, or has not acted on within its jurisdictional authority, and which the peer-reviewed literature associates with population-level increases in neurodevelopmental, immune, metabolic, and other health outcomes that contribute to the prevalence of disability among Alberta residents.

The pillar proceeds in four parts. Section II(b) presents the Alberta-specific contamination data: documented PFAS hotspots on the Federal Contaminated Sites Inventory, pesticide monitoring data from Alberta surface waters, the legacy asbestos and lead-paint burden in the Alberta built environment, the municipal biosolids-to-agricultural-land pathway, and the consumer-waste-disposal pathway through which hazardous consumer products enter Alberta landfills as unsegregated household waste. Section II(c) presents the peer-reviewed developmental and health-outcome literature that associates these classes of exposure with effects of public-health concern, with particular attention to Canadian birth-cohort studies. Section II(d) examines the regulatory gap at both the federal and provincial levels — the documented divergence between what is known about these substances and what is currently in force or in active regulatory development. Section II(e) traces, in a final passage, how the documented exposures enter and persist across a single human life from prenatal development through adulthood.

A note on epistemic discipline: this pillar establishes that documented exposures occur in Alberta, that those exposures are associated at the population level with developmental and health outcomes contributing to disability prevalence, and that the regulatory response to the exposures has been incomplete or delayed. It does not, and cannot, establish that any individual Alberta resident's disability was caused by any specific exposure. The distinction between population-level association and individual-level causation is a fundamental one in epidemiology and toxicology, and the brief observes it throughout. Where a finding is an association rather than a demonstrated causal mechanism, this pillar says so. Where an animal model finding has not yet been confirmed in humans, this pillar says so. Where a study is foundational but limited (sample size, geographic scope, follow-up window), this pillar says so. The cumulative evidentiary weight of the pillar derives not from any single study but from the convergence of multiple lines of evidence across exposure classes, and from the documented presence of those exposures in Alberta specifically.

b. Alberta-Specific Contamination

b.i. Per- and Polyfluoroalkyl Substances (PFAS)

Per- and polyfluoroalkyl substances comprise a class of approximately 15,000 synthetic chemicals characterized by a carbon-fluorine bond that is highly resistant to environmental and metabolic

degradation (Government of Canada, *Draft State of Per- and Polyfluoroalkyl Substances (PFAS) Report*, 2023). The persistence of the perfluoroalkyl moiety is such that the substances have been termed "forever chemicals" in both regulatory and scientific literature. PFAS have been used in industrial and consumer applications since the 1940s, including in firefighting foams (aqueous film-forming foam, or AFFF), grease-resistant food packaging, stain- and water-repellent textile treatments, non-stick cookware coatings, and a wide range of additional applications.

Alberta hosts documented PFAS contamination at multiple federal sites identified in the Federal Contaminated Sites Inventory and reported through Canada's draft and updated *State of PFAS Report* publications by Environment and Climate Change Canada. The Government of Canada's draft report, current to April 2022, identified eight PFAS-contaminated sites in Alberta — four classified as active and four as suspected — alongside additional sites in every other province and territory, the majority associated with past or current use of AFFF in firefighting training and equipment maintenance at airports and military installations (Environment and Climate Change Canada, 2023, *Draft State of PFAS Report*, Figure 3 and accompanying text).

Among the Alberta-specific sites, the following are confirmed by federal documentation:

- **Calgary, Alberta:** Former Firefighting Training Area, Calgary International Airport. Groundwater contamination under assessment. Population within 1 km: 30. Wells within 1 km confirmed present.
- **Edmonton, Alberta:** Canadian Forces Base Edmonton. Surface water, groundwater, and soil contamination. National Defence has reported sampling 15 wells in the surrounding area, none of which exceeded the Health Canada provisional drinking-water guideline of 30 ng/L. Population within 1 km: 87.
- **Banff National Park, Alberta:** Peyto Pit. Groundwater and soil contamination under assessment. No known wells within 1 km.
- **Jasper, Alberta:** Additional federal site identified.

(Source: Federal Contaminated Sites Inventory, as reported in CBC News interactive map of Canadian PFAS hotspots, May 2025, drawing on data current to March 19, 2025; underlying data verifiable through the Federal Contaminated Sites Inventory.)

The figure of 80 federal sites mapped by CBC, drawn from Environment and Climate Change Canada's reporting, is, in the publication's own framing, "likely a vast undercount." Provincial and municipal PFAS monitoring is not standardized across Canada, and the federal map excludes sites under provincial, municipal, and private authority. Université de Montréal toxicologist Marc-André Verner, in commentary published with the CBC reporting, stated that emerging science suggests there may be no safe level of PFAS exposure.

It should be noted that AFFF use was historically widespread at Canadian Forces installations across western Canada, including bases beyond those currently confirmed on the federal inventory. Civil society organizations including the Alberta Wilderness Association have published additional concerns regarding suspected PFAS contamination at additional Alberta sites; those

claims are noted here without independent verification, as the federal inventory remains the authoritative source for confirmed contamination at this writing.

A second pathway for PFAS entry into the Alberta environment is the municipal biosolids-to-agricultural-land application program. The City of Calgary operates the Calgro biosolids land-application program, which has been in continuous operation since 1983. Under this program, biosolids generated by the Bonnybrook Wastewater Treatment Plant are subsurface-injected at a depth of 5-10 cm below the ground surface on pre-approved agricultural land in southeast Calgary, near the Shepard Lagoons, between April and late October each year (City of Calgary, *Biosolids Management Program*, ongoing). Approved crop categories on Calgro-treated land include cereal grains, small oilseeds, dried legumes, forage crops, trees, and sod. The Government of Alberta's 2001 *Guidelines for the Application of Municipal Wastewater Sludges to Agricultural Lands*, still in operative reference at the provincial level, notes that "in Alberta, it is common practice to apply sludge or biosolids to agricultural lands as a nutrient source."

The Canadian Food Inspection Agency, in a notice to industry dated May 2023, confirmed that "there are currently no standards for PFAS in biosolids in Canada." An interim standard of 50 ppb PFOS was proposed for industry consultation; that proposal has not been promulgated as a binding standard at this writing. The first comprehensive Canadian study of PFAS concentrations in biosolids and composts (published in *Science of the Total Environment*, 2024, Quebec sample, n=118 sewage sludge and biosolid samples collected 2019-2023) reported PFAS concentrations of 12 to 1,310 µg/kg in biosolids samples, with diPAPs, sulfonamides, PFCAs, and PFSAAs as dominant chemical classes. Plant uptake of PFAS — particularly PFOA and PFOS — has been documented across at least 16 field studies summarized in the Interstate Technology and Regulatory Council's 2022 fact sheet on biosolids and PFAS, and livestock fed silage from PFAS-contaminated fields have been documented to accumulate PFAS in tissues (Lindstrom et al., 2011; Skutlarek, Exner, and Farber, 2006).

The Calgary configuration is therefore as follows: a federally documented PFAS-contaminated site (Calgary International Airport, former AFFF firefighting training area) lies within the same metropolitan region as a long-running municipal biosolids land-application program that distributes wastewater-derived solids to agricultural land used for human food production, in a regulatory environment in which Canada has not yet established binding standards for PFAS in biosolids and Alberta has not implemented PFAS-specific monitoring of either the biosolids stream or the receiving agricultural soil. The pathway from federal-site contamination, through municipal wastewater treatment that does not remove PFAS, into agricultural soil, into crops and livestock, and into the human food supply, is documented in each of its segments by the responsible public authorities. Calgary does not publicly report PFAS testing of Calgro biosolids at this writing.

b.ii. Pesticides in Surface Water

Alberta Environment has measured pesticide concentrations in provincial surface waters since the mid-1980s. The most comprehensive provincial overview, *Overview of Pesticide Data in Alberta*

Surface Waters Since 1995 (Anderson, Alberta Environment, covering 1995-2002 data, published 2005), reported that of 63 pesticide compounds measured across 35 sampling programs, 44 were detected in Alberta surface waters. Provincially, 2,4-dichlorophenoxyacetic acid (2,4-D) was the most frequently detected pesticide, present in 53% of samples. Seven additional compounds were detected in 10% to 50% of samples (clopyralid, dicamba, glyphosate, MCPA, MCPP, picloram, and triclopyr), and 15 additional compounds were detected in 1% to 10% of samples.

A 2019 synoptic survey of southern Alberta watersheds — conducted by Environment and Climate Change Canada researchers and published in the *Journal of Limnology* (full citation: Glozier et al., 2019, "From peaks to prairies: a time-of-travel synoptic survey of pesticides in watersheds of southern Alberta, Canada," DOI: 10.1080/20442041.2019.1634947) — confirmed the persistence of 2,4-D as the highest-detection-frequency pesticide in surface waters across all studied river subbasins (Oldman, South Saskatchewan, Bow, and Red Deer). The study analyzed 106 pesticide compounds in grab samples from waterways, wastewater treatment plant effluents, and irrigation return flows in Alberta's most heavily farmed regions. Pesticide detection occurred in wastewater treatment plant effluents in both summer (August 2014) and winter (February 2015) sampling, indicating sources beyond seasonal agricultural application.

The 23 watersheds monitored under Alberta's AESA (Alberta Environmentally Sustainable Agriculture) Water Quality Monitoring Project documented detectable pesticide concentrations in every watershed studied, with 37 of 68 monitored compounds detected and a positive correlation observed between agricultural land-use intensity and pesticide detection frequency, total pesticide concentration, and number of compounds detected per sample (AESA Water Quality Monitoring Project, Volume 3, 1999-2006 data).

b.iii. Asbestos

Asbestos is a naturally occurring silicate mineral widely used across North American construction between approximately 1940 and 1990 for its fire-resistance, tensile strength, and thermal insulation properties. Peak use occurred from 1940 through 1975, with continued application in specific products through the 1980s and beyond. Asbestos was incorporated into an estimated three thousand distinct building and industrial products during this period, including spray-applied fireproofing, thermal insulation on pipes and boilers, drywall joint compound, floor tiles, ceiling tiles, cement products (pipes, boards, siding), roofing materials, plaster, gaskets, and brake pads (Government of Canada, *Asbestos and Your Health*, ongoing).

Canadian regulation of asbestos proceeded in incremental stages across more than four decades. A 1976 restriction limited asbestos use in insulation and fireproofing applications. The *Asbestos Mines and Mills Release Regulations* (1977) limited emissions from mining and processing operations. The *Asbestos Products Regulations* (1986) under the *Hazardous Products Act* further restricted asbestos-containing product manufacture and labelling. Asbestos drywall joint compound was banned in 1980. The *Hazardous Products (Crocidolite Asbestos) Regulations* (1989) prohibited the most toxic asbestos variant, and spray-applied asbestos products were fully banned under the *Hazardous Products Act* the same year. Despite these incremental restrictions,

asbestos mining continued in Canada until 2011-2012, when the last active operations in Thetford Mines and the town of Asbestos, Quebec, ceased production. In 2015, Canada was still importing approximately \$8.2 million in asbestos-containing products, primarily brake pads and cement products. The comprehensive ban — *Prohibition of Asbestos and Products Containing Asbestos Regulations* (SOR/2018-196) — came into force on December 30, 2018, forty-two years after the first restriction in 1976. The 2018 regulations include time-limited exemptions for the military, nuclear, and chlor-alkali industries.

The federal Department of National Defence maintains a National Asbestos Inventory documenting which buildings at which Canadian Forces installations contain asbestos-containing materials. Military buildings constructed during the peak use period (1940-1975) are presumptively contaminated unless documented otherwise. Canadian Forces Base Edmonton and other historical military installations in Alberta are within the inventory's scope. Permanent Married Quarters (PMQ) housing on Canadian Forces bases during the relevant construction periods typically incorporated multiple asbestos-containing materials in fireproofing, insulation, and ceiling-tile applications.

In Alberta, the school building stock constructed during the peak asbestos use period contains documented asbestos-containing materials. The Government of Alberta's *Alberta Asbestos Abatement Manual* (2019, and predecessor documents) acknowledges the widespread presence of asbestos in Alberta's pre-1990 institutional, educational, and residential building stock, and establishes the regulatory framework under which known asbestos-containing materials are managed. The current regulatory approach — consistent across Canadian provinces — is management in place rather than universal removal. Under this approach, asbestos-containing materials that remain undisturbed and in good condition are permitted to remain in buildings, with disturbance triggering notifiable abatement procedures under provincial occupational health and safety regulations.

This approach means that Alberta children attending pre-1990 school buildings are, in the ordinary course, not exposed to airborne asbestos at concentrations of acute concern. It also means that the building stock contamination has not been eliminated; it has been deferred. Buildings constructed between 1940 and 1990 that remain in institutional use continue to contain the asbestos-containing materials they were built with. Deterioration of those materials with age, ventilation system disturbance, maintenance activity, and renovation work all represent pathways by which previously encapsulated asbestos can become airborne.

The disease-latency characteristics of asbestos exposure are central to the life-course relevance of this contamination class. Peer-reviewed literature establishes that asbestos-related diseases — asbestosis, mesothelioma, and certain lung cancers — have latency periods ranging from approximately ten to fifty years. Asbestosis typically manifests ten to forty years after first exposure. Mesothelioma typically manifests twenty to fifty years after first exposure. Asbestos-related lung cancer typically manifests fifteen to thirty-five years after first exposure. Higher-intensity and longer-duration exposures tend to produce shorter latency periods and more severe outcomes. Children with early-life exposure typically demonstrate longer latency periods, and a

portion of population-level asbestos disease burden at any given time reflects cumulative exposure in previous decades.

For Alberta's present-day disability population, this latency profile has two structural consequences. First, the adults presently assessed under AISH and transitioning to ADAP include cohorts who were children during the peak Canadian asbestos use period and who may have encountered significant early-life exposure in school, residential, or military-adjacent environments. Second, Alberta children currently attending pre-1990 school buildings are being exposed to managed-in-place asbestos-containing materials under an ongoing regulatory framework that defers rather than eliminates exposure, with symptoms from that exposure anticipated to emerge across the 2040-2070 window.

b.iv. Lead Paint

Lead-based paint was manufactured, sold, and applied in Canada for more than a century prior to effective regulatory restriction. Before 1960, lead was a principal component of Canadian paints, with formulations containing as much as 50% lead by dry weight to provide colour, durability, and faster drying. The neurotoxic effects of lead, particularly on cognitive development in children, have been scientifically documented since the early twentieth century and are among the most thoroughly established findings in environmental health science. The World Health Organization's established position, reiterated in the agency's ongoing guidance, is that there is no known safe level of lead exposure for children.

The Canadian regulatory history of lead in paint is one of incremental restriction across decades, and of regulatory stringency substantially below the contemporaneous United States standard. In 1976, Canada introduced the first restriction on lead in interior paint: a limit of 5,000 parts per million (0.5% by weight). This limit was more than eight times less stringent than the United States standard of 600 parts per million established in 1978. Exterior paints could continue to contain higher lead concentrations through the 1970s and 1980s provided they carried a warning label. Canadian paint manufacturers voluntarily began conforming to near-background lead levels in their interior and exterior consumer paints in 1991 under a voluntary agreement with the federal government.

Full legislative regulation of lead in paint did not come into force in Canada until 2005, when the *Surface Coating Materials Regulations* under the *Canada Consumer Product Safety Act* reduced the permissible lead limit to background levels for both interior and exterior consumer paints. The *Surface Coating Materials Regulation* SOR/2016-193 is the current controlling instrument. Since 2010, any paint sold in Canada containing more than 90 parts per million lead must be labelled to warn against use in areas accessible to children or pregnant women.

Twenty-nine years elapsed between the first Canadian restriction (1976) and the full legislative regulation (2005). During the intervening period, lead-based paint continued to be manufactured, sold, and applied in Canadian homes, schools, institutional buildings, and infrastructure. The 2007 report by the Canadian arm of the Association of Community Organizations for Reform Now (ACORN) estimated that approximately three million Canadian homes — housing approximately

one-quarter of Canadian children — had been built before 1960, when lead paint use was most widespread. Approximately 75% of Canadian homes built before 1978 contain some lead-based paint.

Canada's regulatory approach to lead in paint has been assessed by peer-reviewed public-health scholarship as substantially weaker than the contemporaneous United States framework. Health Canada's historical position — documented in the academic literature as "reformulating lead-based paint as a problem of the past, or as a US problem" — has not corresponded to the Canadian clinical record. The *Canadian Medical Association Journal* documented, in a 2004 case report, a four-year-old Ottawa boy hospitalized for lead poisoning after ingesting paint chips containing 0.5% lead — the lead concentration Canadian federal regulators had designated as acceptable in the 1976 restriction. A 1995 investigation of London, Ontario children (aged 0-17, n=164; Langlois et al., *Canadian Journal of Public Health*, 1996) identified a 7.3% prevalence of elevated blood lead levels (≥ 10 µg/dL), with lead-based paint implicated as the primary source and with children living in pre-1945 homes demonstrating average blood lead levels 62.3% higher than those living in later-built homes ($p=0.01$).

For Alberta, the lead paint contamination profile mirrors the national pattern. Alberta housing constructed before 1991, and particularly before 1978, contains lead paint at rates broadly consistent with the national estimates. Alberta school buildings constructed during the relevant periods contain documented lead paint, subject to the same "manage in place" regulatory approach as asbestos. Lower-income Alberta households — including the AISH and ADAP recipient population examined in Pillar III — are disproportionately housed in older rental housing stock less likely to have been remediated through subsequent renovation investment. The documented correlation between pre-1945 housing age and elevated childhood blood lead levels observed in the Ontario investigation cited above is generalizable to comparable Alberta housing stock. The on-reserve housing stock in Alberta First Nations communities, which falls under federal jurisdiction, is characterized by older construction, deferred maintenance, and documented lead exposure risks, as examined in the campaign's separate Indigenous Disability evidentiary report.

The developmental consequences of childhood lead exposure are uniquely well-established among the environmental contamination classes this pillar examines. The scientific consensus, supported by decades of cohort research and reviewed in the positions of the World Health Organization and the Centers for Disease Control and Prevention, is that there is no known safe level of lead exposure for children. Cognitive, behavioural, and neurodevelopmental outcomes associated with elevated childhood blood lead levels include reductions in measured IQ, increased incidence of attention-deficit and hyperactivity disorders, reduced academic achievement, and increased behavioural difficulty. These outcomes directly overlap with the diagnostic profiles that result in Alberta Education special-education coding (particularly Codes 41, 42, 43, and 44), with eligibility for Family Support for Children with Disabilities (FSCD) supports, and with subsequent adult eligibility for Persons with Developmental Disabilities (PDD) and AISH/ADAP income supports.

The MIREC Canadian birth cohort has been used to examine prenatal lead exposure and cognitive outcomes in Canadian children, with results consistent with the broader international literature on

the developmental effects of low-level prenatal lead exposure. Full MIREC lead-specific citations will be compiled in Appendix B.

Like asbestos, the lead paint contamination burden in Alberta has an inter-generational character. The adult AISH population includes cohorts exposed to lead paint during childhood in the 1960s through 1980s, prior to meaningful regulatory restriction. The children currently presenting with neurodevelopmental concerns for FSCD assessment include cohorts with ongoing exposure to managed-in-place lead paint in older housing and school buildings. The same population, across generations, is bearing the developmental consequences of a regulatory framework that has, in the case of lead paint, repeatedly deferred abatement in preference to labelling, warning, and management-in-place approaches.

b.v. Consumer Waste Disposal and Landfill Leachate Pathways

In addition to the federal contaminated-site inventory and the municipal biosolids-to-agricultural-land pathway discussed in Section II(b)(i), a third documented pathway by which PFAS and other hazardous consumer-product constituents enter the Alberta environment is the disposal of household waste into municipal landfills. This pathway is structurally distinct from the previous two in that it arises not from industrial contamination or from the wastewater treatment loop, but from the ordinary, legally permitted disposal of consumer products by Alberta households into the regular residential waste stream.

Alberta's *Waste Control Regulation* (Alta Reg 192/1996), promulgated under the *Environmental Protection and Enhancement Act*, is the province's principal regulatory instrument governing waste classification and disposal. Section 3(a) of the Regulation explicitly excludes from hazardous waste classification "household waste in the possession of the householder or while unsegregated in a municipal waste management system." The practical effect of this exclusion is as follows: consumer products containing substances that, if generated as industrial waste, would be classified and managed as hazardous — such as PFAS-coated non-stick cookware, stain-resistant or water-repellent textiles, PFAS-treated grease-resistant food packaging, certain cosmetics and personal-care products, electronic components, and products containing residual lead, cadmium, or mercury — are not, under Alberta law, treated as hazardous waste when disposed of by consumers in the normal course. They become hazardous waste only if separated from the unsegregated household stream, typically through municipal Household Hazardous Waste (HHW) programs or time-limited Toxic Roundup events. While they remain in the household waste stream, the regulatory framework treats them as ordinary municipal solid waste.

Alberta's landfill system, regulated under the *Waste Control Regulation* and the *Standards for Landfills in Alberta*, operates in three classes. Class I landfills are authorized to accept certain specified hazardous wastes at specified concentration limits (for example, solid waste containing halogenated organic compounds at combined concentrations below 1,000 mg/kg, with polychlorinated biphenyls below 50 mg/kg). Class II landfills accept waste that does not include hazardous waste; the majority of Alberta residential garbage is directed to Class II facilities. Class III landfills accept only inert waste. Consumer products containing PFAS, regulated at the federal

level as toxic under CEPA as described in Section II(d)(i), routinely enter Class II landfills through household disposal.

Peer-reviewed environmental engineering literature establishes that landfill leachate — the liquid that forms as precipitation and moisture move through deposited waste — is a significant release pathway for PFAS from landfilled consumer products to surrounding soil, surface water, and groundwater. Lang et al. (2017), "National Estimate of Per- and Polyfluoroalkyl Substance (PFAS) Release to U.S. Municipal Landfill Leachate," *Environmental Science & Technology* 51(4): 2197-2205, reported PFAS detection in 95 of 95 landfill leachate samples analyzed across 18 U.S. landfills, with total PFAS concentrations in leachate ranging across three orders of magnitude. Canadian landfill leachate studies have reported consistent findings. The concentrations of PFAS released per tonne of municipal solid waste are sufficient, in aggregate across the North American landfill system, to represent a principal ongoing source of environmental PFAS release.

Municipal Household Hazardous Waste programs in Alberta are voluntary at both the municipal and household levels. The Government of Alberta's official guidance to residents is to "check with your local municipality to determine if they have a household hazardous waste collection site or if they host temporary round-up events." The City of Calgary operates a free HHW drop-off program with eleven locations (eight fire-station drop-off sites and three Eco Centres at municipal landfills) and provides for the free disposal of solid and liquid hazardous products bearing the federal Consumer Chemical and Containers Regulations (CCCR) hazard symbol. The City of Edmonton operates Eco Stations. Many of Alberta's smaller urban municipalities, rural municipalities, and Indigenous communities operate seasonal roundups, limited drop-off facilities, or, in some cases, no dedicated HHW infrastructure at all. There is no provincial requirement that consumers separate hazardous products from their regular household waste, and no provincial penalty for failing to do so.

On April 1, 2025, Alberta launched an Extended Producer Responsibility (EPR) program, overseen by the Alberta Recycling Management Authority (ARMA), covering three product categories: single-use products, packaging and printed paper (PPP), and hazardous and special products (HSP). EPR shifts financial responsibility for end-of-life management of designated products from municipalities to the producers of those products. The HSP category incorporates some materials previously handled under the pre-existing HHW framework. The 2025 EPR designated materials list, however, does not include most PFAS-containing consumer products (including non-stick cookware, PFAS-coated textiles, and PFAS-treated food packaging), and does not include asbestos-containing consumer products or lead paint residuals. The EPR framework is narrow in its current scope and is structured around producer financial responsibility rather than mandatory consumer sorting or diversion.

The regulatory configuration that results from these provisions is as follows: a consumer product containing a substance classified as toxic under federal legislation, purchased at retail in Alberta, used by an Alberta household, and discarded into the household waste stream, will in the ordinary course be deposited into a Class II landfill not designed for hazardous-waste containment, without having been classified or handled as hazardous at any point in its life cycle under Alberta law. The

Alberta-specific structural gap this produces, in combination with the federal contaminated-site inventory and the municipal biosolids-to-agricultural-land loop, is that three distinct pathways — none of which is, at this writing, the subject of active Alberta-specific regulatory remediation — concurrently contribute to the ongoing release of PFAS and other legacy contaminants into the Alberta environment.

b.vi. The Indigenous Disability Context

The Alberta Disability System Breakdown Campaign has produced separate evidentiary documentation on the specific intersection of environmental contamination, disability, and Indigenous communities in Alberta — the *Indigenous Disability in Alberta: A 195-Page Evidentiary Report*, available through the campaign's distribution channels and submitted to the Yellowhead Institute, the United Nations Special Rapporteur on the Rights of Persons with Disabilities, and other relevant bodies. That document examines the historical and present-tense conditions — contaminated land and water on or near reserves, jurisdictional fracture in service delivery, the Mortality Gap, the specific application of TRC Calls to Action 18-24 on health, the operation of Jordan's Principle, and related considerations — that together produce documented disability rates among Indigenous Albertans substantially in excess of the non-Indigenous population. The framework established in that document, particularly with respect to the relationship between environmental conditions and population-level disability outcomes, is foundational to the present pillar's extension of analogous reasoning to the broader Alberta population.

Readers concerned with the Indigenous-specific dimensions of the compound pattern this brief examines are directed to the Indigenous Disability evidentiary report as a primary source on those dimensions. The present pillar does not attempt to summarize that work; it cites it as established framework and proceeds.

c. Peer-Reviewed Developmental and Health-Outcome Literature

c.i. Prenatal PFAS Exposure and Childhood Cognitive Outcomes

The most directly applicable Canadian primary source on prenatal PFAS exposure and childhood cognitive outcomes is Goodman et al. (2023), "Prenatal exposure to legacy PFAS and neurodevelopment in preschool-aged Canadian children: The MIREC cohort," *Neurotoxicology and Teratology*, Volume 98, Article 107181, DOI: 10.1016/j.ntt.2023.107181 (published online 11 May 2023). The study analyzed first-trimester maternal plasma concentrations of three legacy PFAS compounds — perfluorooctanoic acid (PFOA), perfluorooctanesulfonic acid (PFOS), and perfluorohexanesulfonic acid (PFHxS) — measured in participants of the Maternal-Infant Research on Environmental Chemicals (MIREC) study, a multi-site Canadian pregnancy and birth cohort recruited between 2008 and 2011 across ten Canadian cities, including Edmonton, Alberta. Children's intelligence scores were assessed at ages three to four years using the Wechsler Preschool and Primary Scale of Intelligence, Third Edition (WPPSI-III), yielding Full-Scale IQ (FSIQ), Performance IQ (PIQ), and Verbal IQ (VIQ) measures.

For the analyses reported, n = 522 mother-child pairs were included for FSIQ assessment, n = 517 for PIQ, and n = 519 for VIQ. The authors employed both single-pollutant multiple linear regression and weighted quantile sum (WQS) regression to assess the impact of the PFAS mixture as a whole.

The principal findings of public-health relevance are as follows: every quartile increase in the WQS index of the PFAS mixture was associated with a decrease of 3.16 points in performance IQ in male children ($\beta = -3.16$, 95% CI: -4.90, -1.43). PFHxS contributed the largest weight to the WQS index for this association. No statistically significant association was observed for female children ($\beta = 0.63$, 95% CI: -0.99, 2.26). No statistically significant associations were observed between PFAS exposure and executive function measures (working memory, planning/organizing) in either sex. The single-pollutant analyses produced consistent patterns: higher individual PFAS concentrations were associated with lower performance IQ in male children, with PFHxS as the chemical of consistent concern across analytic approaches.

The authors note that subtle alterations in brain development can have wide-ranging impacts at both the individual and population level, citing Lanphear (2015). The MIREC cohort is, by the authors' own acknowledgment, somewhat older, predominantly Caucasian, and of higher household income and educational attainment than the general Canadian birthing population (Arbuckle et al., 2013); the directionality of any bias this introduces with respect to PFAS exposure-outcome relationships is not straightforward to predict.

The Goodman et al. (2023) paper is co-authored by researchers from, among other institutions, the Department of Pediatrics, Faculty of Medicine and Dentistry, University of Alberta, providing direct Alberta institutional involvement in the foundational Canadian primary research on this question.

c.ii. Ultraprocessed Food Consumption and Behavioral Outcomes

A second Canadian primary source addresses dietary exposure rather than chemical contamination, and is included in this pillar because the underlying economic conditions of Alberta disability income recipients, examined in Pillar III, structurally constrain dietary access in ways that have measurable downstream effects on the developmental outcomes of children in those households.

Kavanagh et al. (2026), "Ultraprocessed Food Consumption and Behavioral Outcomes in Canadian Children," *JAMA Network Open*, Volume 9, Issue 3, Article e260434, DOI: 10.1001/jamanetworkopen.2026.0434 (published 3 March 2026), reports findings from a cohort study examining the association between intake of ultra-processed foods (UPF) at age three years and behavioral outcomes at age five years among children participating in the CHILD Cohort Study in Canada. The analytic sample comprised over 2,000 children. Behavioral outcomes were measured using the Child Behavior Checklist, a validated parent-report instrument.

The authors found that for every 10% increase in calories from ultra-processed foods at age three, children at age five demonstrated higher scores on internalizing behaviour subscales (anxiety, fearfulness), externalizing behaviour subscales (aggression, hyperactivity), and overall

behavioural difficulty composites. Sugar-sweetened beverages and artificially sweetened drinks demonstrated stronger associations with adverse behavioural outcomes than other UPF categories.

The principal investigator, Kozeta Miliku of the University of Toronto's Temerty Faculty of Medicine, observed that the preschool years are critical for child development and that the findings underscore the need for early-life interventions, public health campaigns, nutrition standards for child-care providers, and reformulation of packaged foods. The authorship of the study includes researchers from the Department of Pediatrics, Faculty of Medicine and Dentistry, University of Alberta — providing, again, direct Alberta institutional involvement in the foundational Canadian primary research on this question.

c.iii. Pesticide Exposure and Childhood Neurodevelopment

The MIREC cohort has also been used to examine prenatal exposure to organophosphate pesticides and IQ scores in Canadian children. Ntantu Nkinsa et al. (2020), "Organophosphate pesticides exposure during fetal development and IQ scores in 3 and 4-year old Canadian children," *Environmental Research*, Volume 190, Article 110023, DOI: 10.1016/j.envres.2020.110023, examined first-trimester maternal urinary metabolites of organophosphate pesticides (three dimethyl alkyl phosphate metabolites and three diethyl alkyl phosphate metabolites) in MIREC participants, with WPPSI-III IQ assessment of children at ages three to four years. Sex-specific analyses were performed in light of prior literature reporting more pronounced cognitive deficits in male children with prenatal organophosphate exposure (Marks et al., 2010; Horton et al., 2012; Liu et al., 2016; Wang et al., 2017).

The MIREC cohort has further been used to examine prenatal lead exposure (Ntantu Nkinsa et al., 2020, *Environmental Research*, Volume 190, Article 110023) and prenatal fluoride exposure (Green et al., 2019, "Association Between Maternal Fluoride Exposure During Pregnancy and IQ Scores in Offspring in Canada," *JAMA Pediatrics*, Volume 173, Issue 10, pages 940-948), with consistent findings of sex-specific cognitive effects in male children at the population level.

The convergence across exposure classes — PFAS, organophosphate pesticides, lead, fluoride — of similar sex-specific findings (greater male susceptibility to prenatal exposure-related cognitive effects) in the same Canadian cohort, using the same standardized cognitive assessment instrument, with overlapping authorship and methodology, strengthens the population-level inference even where any single exposure class would not, on its own, support a categorical claim. The MIREC lead findings are of particular relevance given the ongoing presence of lead-based paint in the Alberta built environment, as examined in Section II(b)(iv) above: residential and institutional lead paint represents a documented, non-dietary, non-occupational source of ongoing childhood lead exposure in the affected population.

c.iv. PFAS Health Effects in the Broader Literature

Beyond the Canadian birth-cohort literature, the broader peer-reviewed literature on PFAS health effects in humans includes documented or strongly suspected associations with: reduced birth weight; reduced vaccine antibody response in children (Grandjean et al., 2012, and subsequent

replications); thyroid disease; hypercholesterolemia; ulcerative colitis; kidney cancer; and testicular cancer. The C8 Health Project — a court-ordered epidemiological study of approximately 69,000 residents of the Mid-Ohio Valley region with documented historical PFOA exposure from a 3M/DuPont manufacturing facility — established what the C8 Science Panel termed "probable links" between PFOA exposure and six health conditions: kidney cancer, testicular cancer, thyroid disease, ulcerative colitis, pregnancy-induced hypertension, and high cholesterol.

PFAS cross the placental barrier; placental transfer is documented and the strength of the maternal-cord blood concentration correlation is well-established in the Canadian literature (Fisher et al., 2016, and subsequent MIREC analyses; in MIREC, the percentage of maternal plasma samples with detectable PFOA, PFOS, and PFHxS was 99.8%, 99.8%, and 95.7% respectively). PFAS are excreted in breast milk and represent a documented exposure pathway from the lactating parent to the nursing infant. PFAS are not effectively metabolized by the human body; serum half-lives for the legacy long-chain PFAS range from approximately three to eight years (Fenton et al., 2021).

A growing body of animal-model literature examines transgenerational and epigenetic effects of PFAS exposure, including effects on germ-cell DNA integrity, transgenerational alterations in gene expression patterns regulating neurodevelopment and thyroid function, and effects on offspring developmental outcomes in animals not directly exposed. A representative recent example is the University of Adelaide research on low-level PFAS exposure in drinking water and embryo development (April 2026). These findings have not at this writing been confirmed in human longitudinal studies; the brief notes them as suggestive and cites them as pointers to further research, not as established findings in human populations.

d. The Regulatory Gap

This section examines the regulatory gap at the federal and provincial levels for the contamination classes introduced in Sections II(b)(i) and II(b)(ii). The regulatory histories of asbestos and lead paint are self-contained within Sections II(b)(iii) and II(b)(iv) above, and the consumer-waste and landfill-leachate regulatory framework is self-contained within Section II(b)(v); these are not duplicated here. The pillar's overall regulatory-gap findings include those subsections' documented histories of decades-long delay between first restriction and full regulation, and the documented provisions of Alberta's current waste-classification regime.

d.i. Federal Regulation of PFAS

The Government of Canada published its draft *State of Per- and Polyfluoroalkyl Substances (PFAS) Report* in May 2023, with an updated version in 2024 and a finalized version in March 2025 (Environment and Climate Change Canada and Health Canada, *Updated Draft State of PFAS Report* and *State of PFAS Report*). The report concluded, on the basis of evidence reviewed, that PFAS as a class meet the criteria for designation as toxic substances under the Canadian Environmental Protection Act, 1999 (CEPA). On the basis of the State of PFAS Report, the federal

government published an associated *Risk Management Approach for Per- and Polyfluoroalkyl Substances (PFAS)* in March 2025 outlining proposed restrictions.

The actual implementation timeline of binding regulations under this approach extends consultation processes into 2027 and beyond, with phased restrictions on PFAS in firefighting foams (the *Prohibition of Certain Toxic Substances Regulations, 2025*, published 31 December 2025, coming into force 30 June 2026 with respect to AFFF specifically), in cosmetics, in food packaging, and in additional consumer-product categories proposed for subsequent stages. The C8 Aqueous Film-Forming Foam category — the legacy long-chain formulation associated with the bulk of historical contamination at federal sites including those documented in Alberta — is subject to time-limited exemptions extending to the end of 2030 to accommodate transition to alternatives.

Canada has, since 2012, listed three PFAS — PFOS, PFOA, and long-chain perfluorocarboxylic acids — as toxic under CEPA, with regulatory restrictions on each. The remaining approximately 15,000 substances in the PFAS class have, until the 2025 risk management approach, been substantially unregulated.

The Canadian Food Inspection Agency confirmed in May 2023 that no Canadian standards exist for PFAS in biosolids used as commercial fertilizer. A proposed interim standard of 50 ppb PFOS has been advanced for stakeholder consultation but has not been promulgated.

d.ii. Provincial Regulation in Alberta

Alberta has not implemented standalone provincial regulations on PFAS in air, water, soil, or biosolids beyond the federal framework. Alberta's *Environmental Protection and Enhancement Act* (EPEA) provides the statutory architecture for provincial environmental regulation, and the *Tier 1 Soil and Groundwater Remediation Guidelines* and *Tier 2 Soil and Groundwater Remediation Guidelines* published by Alberta Environment and Protected Areas establish remediation criteria for contaminated sites. The application of these criteria to PFAS at provincially-regulated sites is an active area of regulatory development at this writing; the brief notes the absence of finalized province-specific PFAS criteria as a regulatory gap rather than asserting a particular criterion has been violated.

Alberta surface water pesticide monitoring, as described in Section II(b)(ii), is conducted under provincial authority. The most recent comprehensive provincial overview of the data covers the period through 2002. More recent and ongoing monitoring exists under various programs (the AESA Water Quality Monitoring Project, the Long Term River Network, and others), but a current, public, provincially-issued comprehensive overview of pesticide data in Alberta surface waters comparable to the 2005 Anderson publication is not, at this writing, in current circulation. Updated provincial reporting on pesticide presence in Alberta surface waters would meaningfully strengthen the public's ability to assess current exposure conditions.

Provincial advisory infrastructure concerning water and air quality has been further reduced in the same fiscal cycle in which this brief is being drafted. On February 26, 2026, the Alberta Water

Council — a multi-stakeholder advisory body established in 2004 to support implementation of the Government of Alberta's Water for Life strategy (first released in 2003) — received a letter from Alberta Environment and Protected Areas notifying the Council that its provincial grant funding would cease May 26, 2026, with the Council given ninety days to wind up operations and return unused funds to the department. The Council's twenty-three members at the time of the notification included representatives of the Canadian Association of Petroleum Producers, Team Alberta Crops, the Environmental Law Centre, and Ducks Unlimited Canada, among other industry and non-governmental organizations. In the same cycle, the Clean Air Strategic Alliance — Alberta's parallel multi-stakeholder air quality advisory body — also received notice of funding termination. A spokesperson for Alberta Environment and Parks stated that the combined funding decision for both councils would produce annual savings of approximately \$1 million and that "ending funding for these councils does not change Alberta's commitment to maintaining the highest environmental and regulatory standards" (Alberta Environment and Parks statement to CBC News, April 20, 2026). The elimination of the province's two principal multi-stakeholder environmental advisory bodies occurs in the same fiscal year as the provincial regulatory gaps in PFAS monitoring, pesticide surface water reporting, and household hazardous waste handling documented in the preceding subsections, and thirty-five days before the AISH-to-ADAP transition examined in Pillar III takes effect.

d.iii. The Federal-Provincial Jurisdictional Configuration

The PFAS contamination documented in Alberta at Calgary International Airport, Canadian Forces Base Edmonton, and other federal sites is contamination on federally-controlled land. Federal jurisdiction governs the contamination source. Provincial jurisdiction governs human-services responses to Alberta residents — including the disability income, health, and educational support programs examined in Pillars II and III of this brief — affected by the downstream consequences of that contamination. Federal jurisdiction governs the Canada Disability Benefit; provincial jurisdiction governs the choice to claw it back, which Alberta has uniquely exercised.

The brief returns to this jurisdictional configuration in Section V's legal analysis. For the present pillar, the configuration produces the following condition: Alberta children whose prenatal and ongoing exposure to substances on the federal contaminated-sites inventory cannot be regulated by the province, who reach adulthood with conditions that render them eligible for the provincial disability income program, will under ADAP find that program reduced, that program's federal complement clawed back by the province, and that program's independent appeal mechanism removed by the province. The harm, in other words, has a federal source, a developmental pathway, and a provincial endpoint. Each level of government has authority over a different segment of the chain. None of the segments, currently, are being executed in a manner that protects the population the chain affects.

e. The Life Course

The evidence assembled in this pillar describes population-level associations and documented exposures. To the reader keeping in view that the pillar's evidentiary weight derives from

convergence across the full life course, the following passage traces how the documented Alberta exposures enter and persist across a single human life.

Prenatal. PFAS cross the placental barrier. In the MIREC Canadian birth cohort, 95.7% to 99.8% of first-trimester maternal plasma samples contained detectable concentrations of the three legacy PFAS measured. Higher prenatal exposure to the PFAS mixture was associated, in the MIREC cohort, with measurable decreases in performance IQ in male children at age three (Goodman et al., 2023). Organophosphate pesticide metabolites in maternal urine were detectable in the same cohort; analogous sex-specific cognitive effects were reported. Maternal fluoride exposure in fluoridated and partially fluoridated Canadian municipalities was associated, in the same cohort, with lower offspring IQ, particularly in male children (Green et al., 2019). The prenatal period, in other words, is one in which Alberta residents — like other Canadians — are documented to have measurable body burdens of multiple substances each independently associated with cognitive outcomes in offspring.

Early childhood. The persistence of legacy PFAS in human tissue (serum half-lives of three to eight years) means that the prenatal body burden is not cleared in early childhood; it accumulates further through ongoing dietary, water, and consumer-product exposures. Breastfeeding — which the evidence overwhelmingly indicates is the developmentally optimal feeding choice — also represents a documented pathway by which PFAS pass from the lactating parent to the nursing infant. Concurrent exposure to ultraprocessed foods, the prevalence of which in low-income Alberta households is structurally driven by the income inadequacy examined in Pillar III, is associated in Canadian preschoolers with measurable increases in behavioural and emotional difficulties at age five (Kavanagh et al., 2026). Dietary access to whole, minimally processed foods is, at current AISH and forthcoming ADAP rates set against current Alberta food costs, mathematically constrained for the majority of recipients — a finding established in the campaign's earlier *Financial Reality Report* and revisited in Pillar III. The early-childhood period is also the period during which symptoms of developmental concern typically become observable to parents and caregivers, triggering the application process to the Family Support for Children with Disabilities (FSCD) program examined in Pillar II. Inclusion Alberta's January 2025 survey research documents waits of up to three years for full FSCD service access following application, during which critical developmental windows close.

School-age. The Alberta school building stock, and Alberta's older residential housing stock, contain documented asbestos-containing materials and lead-based paint, as described at length in Sections II(b)(iii) and II(b)(iv) above. Under the current "manage in place" regulatory approach, Alberta children attending pre-1990 institutional buildings are exposed, at concentrations below thresholds triggering acute concern, to the accumulated environmental contamination of the construction practices of the mid-twentieth century. Lead paint exposure is of particular relevance to this pillar's argument: the neurodevelopmental effects of childhood lead exposure — reductions in IQ, increased attention and behavioural difficulty, and cognitive delay — map directly onto the diagnostic categories that result in Alberta Education special-education coding, Family Support for Children with Disabilities assessment, and subsequent adult eligibility for the disability income supports examined in Pillar III. School-age is also the period in which children diagnosed with

developmental concerns receive Alberta Education special-education coding (Code 41 through 47), the consequences of which for inclusive education access, classroom support, and individual program planning are examined in Pillar II.

Adult. The accumulated body burden of legacy PFAS, organophosphate pesticide exposure, childhood lead-paint exposure where applicable, and other persistent or recurring environmental exposures from the preceding three life stages does not clear in adulthood. For the cohort exposed in childhood to asbestos-containing materials in pre-1990 buildings, the latency window for asbestos-related disease overlaps with the adult life stage — asbestosis, mesothelioma, and asbestos-related lung cancer manifesting decades after first exposure. The documented adult-onset health outcomes associated with elevated PFAS body burden — kidney cancer, testicular cancer, thyroid disease, ulcerative colitis, hypercholesterolemia, and pregnancy-induced hypertension among others — interact with the cognitive and developmental outcomes that may have been established earlier in the life course through prenatal and childhood lead, pesticide, and other exposures. The adult who arrives, in this stage, at the disability income system examined in Pillar III does so with a cumulative exposure history the science of which is documented and the regulatory response to which has, in each of its elements, been incomplete. The endpoint examined in Pillar III is not the only consequence of the exposures this pillar describes. It is the consequence the brief is principally constructed to examine.